

SANDANIAINA TSINJO NANTOSOA

AI Engineer | Agentic AI | RAG | Python/FastAPI | Full-Stack AI

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PROFESSIONAL SUMMARY

AI Engineer with a product-minded approach to building production AI products: AI agents, RAG systems, and full-stack AI applications end-to-end. Proficient in Python and other programming languages, building applications with FastAPI, LangChain/LangGraph, LLM tool calling, and agent orchestration, backed by Qdrant vector retrieval, React, PostgreSQL, and Redis. Practical background in document ingestion, hybrid retrieval, evaluation, reliability, and monitoring, from architecture through production deployment on Docker/CI-CD.

PROFESSIONAL EXPERIENCE

Humans4Help Madagascar — AI Engineer, Conversational AI & RAG Systems

January 2026 – July 2026 | Antananarivo, Madagascar

- Developed Python and FastAPI backend services for conversational AI, retrieval, business APIs, and external integrations, deployed to production.
- Built production RAG knowledge pipelines: ingestion, chunking, embedding generation, Qdrant indexing, retrieval, and automated weekly refresh (700+ chunks).
- Implemented multi-tenant architecture with tenant-specific knowledge isolation, prompts, configuration, and session management, scalable to 10+ clients.
- Implemented real-time response streaming with Server-Sent Events (SSE) for conversational AI.
- Built reliability mechanisms: input validation, error handling, fallback behavior, rate limiting, structured logging, health checks, and monitoring.
- Implemented security controls: prompt-injection protection, GDPR compliance, PII-aware processing, and audit logging.
- Set up Docker Compose and GitHub Actions CI/CD pipelines for backend testing and automated deployment.

Key results: 3 conversational AI assistants deployed in production for European corporate websites; 97/98 quality checks passed; AI output quality improved 17–20 points; 0 critical errors on security test batches.

Stack: Python, FastAPI, LangChain, OpenAI, Qdrant, PostgreSQL, Redis, Docker, GitHub Actions, SSE, JWT, RBAC.

Freelance — Mathematics & Physics Instructor

October 2019 – Present | Madagascar

- Translate complex technical concepts into clear, structured explanations, directly applied to AI system documentation and communication.
- Develop structured learning materials, applying analytical reasoning and independent problem solving.

SELECTED AI ENGINEERING PROJECTS

AI Business Operating System (AI BOS) — Agentic Business Platform — In Progress

Live Demo: <https://ai-business-os-murex.vercel.app/>

Full-stack AI business platform (CRM, tasks, projects, workflows) with real LLM tool/function calling and agentic execution. Demonstrates agent infrastructure and end-to-end AI product engineering.

- Implemented LLM tool/function calling and multi-step agent execution: the agent selects tools, receives results, and continues until a final response.
- Built agent orchestration with human-in-the-loop approval for sensitive or state-changing actions.
- Added RBAC and permission validation before agents can execute tools or modify business data.
- Implemented AI observability: trace IDs, token usage, cost, latency, and tool execution status monitoring.
- Built with FastAPI, React, PostgreSQL, and Redis, separating deterministic business logic from AI reasoning.

Key areas: agent orchestration, tool calling, human-in-the-loop, RBAC, observability, FastAPI, React, PostgreSQL.

Production Multi-Tenant RAG Platform — Conversational AI & Retrieval System

- Designed a production RAG architecture with document/knowledge ingestion, embeddings, and Qdrant vector retrieval.
- Implemented multi-tenant knowledge isolation with tenant-specific configuration and retrieval.
- Delivered grounded responses via SSE streaming, with reliability and security controls, deployed on Docker.

Key areas: production RAG, Qdrant, multi-tenant architecture, SSE, Docker.

SIHIA — Full-Stack AI Healthcare Platform

Live Demo: <https://sihia-platform.vercel.app/>

Demonstrates end-to-end ownership of an AI feature: ingestion, retrieval, API, frontend, evaluation, testing, and deployment.

- Implemented document ingestion, chunking, and embeddings, indexed into Qdrant for dense retrieval.
- Built hybrid retrieval combining dense search and BM25 with Reciprocal Rank Fusion (RRF), plus cross-encoder reranking.
- Implemented grounded LLM answers with structured source citations and explicit no-answer handling.
- Built a RAG evaluation harness covering retrieval quality, source recall, and generation evaluation.
- Delivered with FastAPI, React, automated testing, and Docker-based deployment.

Key areas: RAG evaluation, hybrid retrieval, reranking, grounded citations, FastAPI, React, Docker.

TECHNICAL SKILLS

AI & LLM Engineering: AI Agents, Agentic AI, Agent Orchestration, Tool/Function Calling, Human-in-the-Loop, RAG, LangChain, LangGraph, OpenAI API, Prompt Grounding

RAG & Information Retrieval: Qdrant, Dense + BM25 Hybrid Retrieval, RRF, Cross-Encoder Reranking, Document Ingestion & Chunking, Source Citations, RAG Evaluation

Backend Engineering: Python, FastAPI, REST APIs, SSE, Pydantic, SQLAlchemy, JWT & RBAC, Rate Limiting, Health Checks

AI Reliability & Observability: LLM Tracing, Token/Cost/Latency Monitoring, Structured Logging, Error Handling & Fallback, Human Approval Workflows, Evaluation, PII-Aware Processing

DevOps & Software Engineering: Docker & Docker Compose, GitHub Actions CI/CD, Linux & Git, Automated Testing (Pytest), Clean Architecture

Frontend & Databases: React, TypeScript, Tailwind CSS, PostgreSQL, Redis, Qdrant, SQL, REST Integration, WebSocket & SSE

AI Development Tools: OpenAI Codex, AI-assisted software development

EDUCATION

Master 2, Applied Mathematics, Computer Science and Statistics (MISA) — University of Antananarivo. Statistics, Machine Learning, Algorithmics, Data. 2026.

Bachelor, Applied Mathematics, Computer Science and Statistics (MISA) — University of Antananarivo. 2024.

CERTIFICATIONS

Associate AI Engineer for Developers — DataCamp:	View certificate
Data Engineer Associate — DataCamp:	View certificate
Data Analyst Associate — DataCamp:	View certificate
Python Data Associate — DataCamp:	View certificate
SQL Associate — DataCamp:	View certificate
Professional Data Analyst — DataCamp:	View certificate

LANGUAGES

Malagasy — Native | French — Fluent | English — Professional Working Proficiency